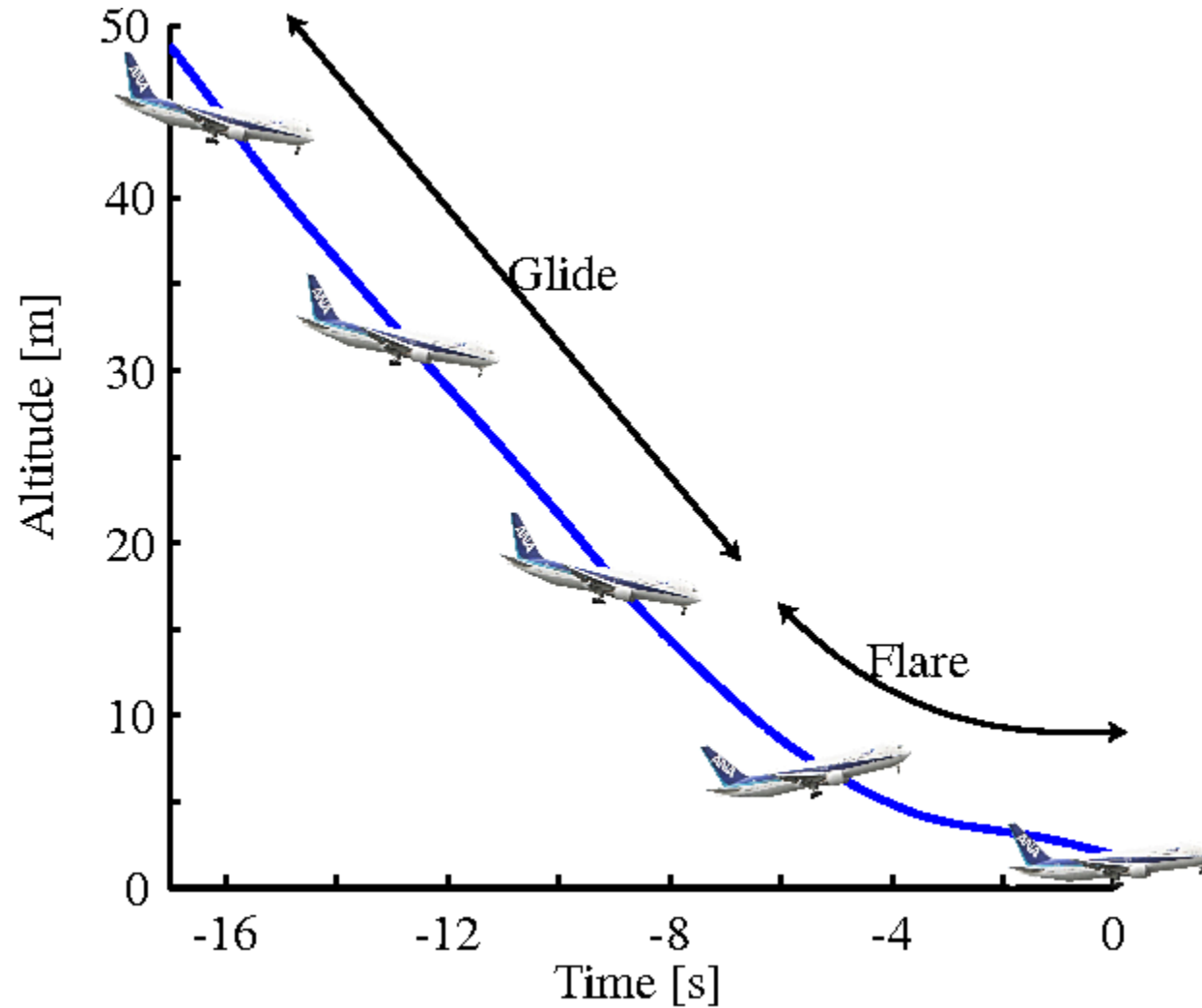


Aircraft Flare Model

MBSE URA2

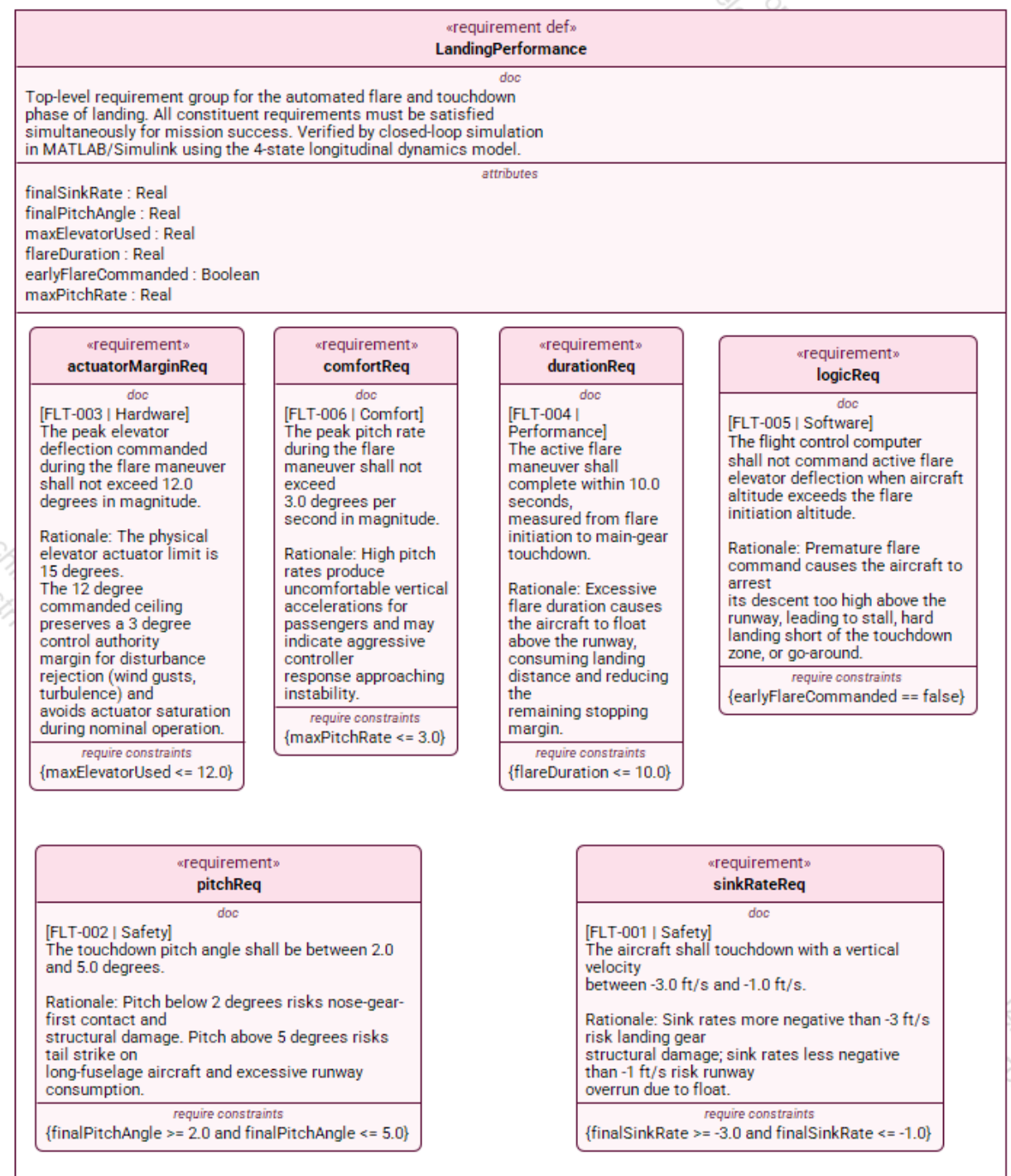
Siddharth Balraj

Landing Flare Maneuver



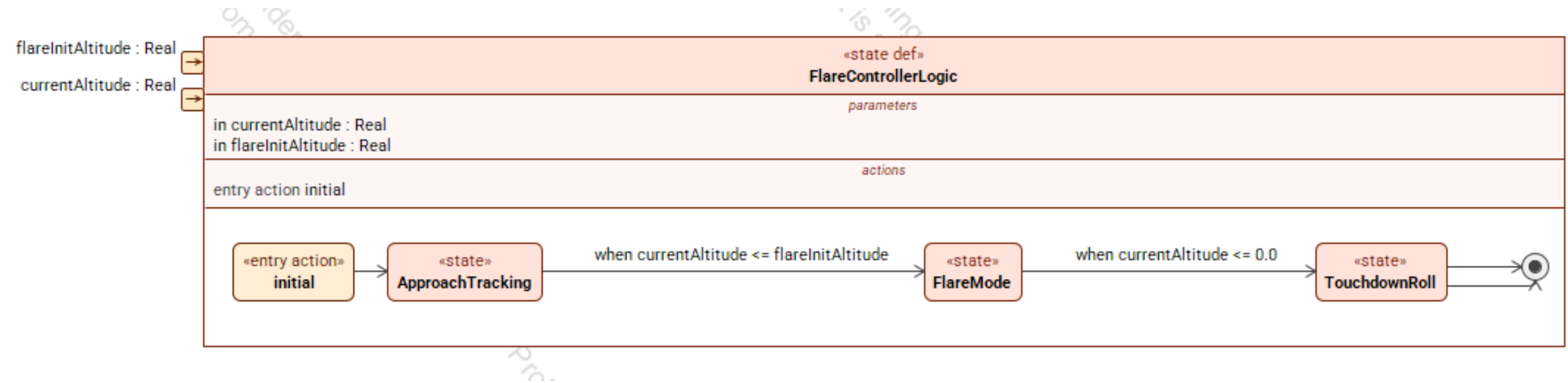
System Requirements

- Sink rate requirement:
 - Vertical velocity between -3.0 ft/s and -1.0 ft/s
- Pitch requirement:
 - Touchdown pitch angle between 2.0 and 5.0 degrees
- Actuator requirement:
 - Peak elevator deflection commanded less than 12 degrees
- Comfort requirement:
 - Peak pitch rate less than 3.0 degrees/second
- Duration requirement:
 - Flare maneuver completed within 10 seconds
- Logic requirement:
 - Flight control computer activates flare only at target altitude

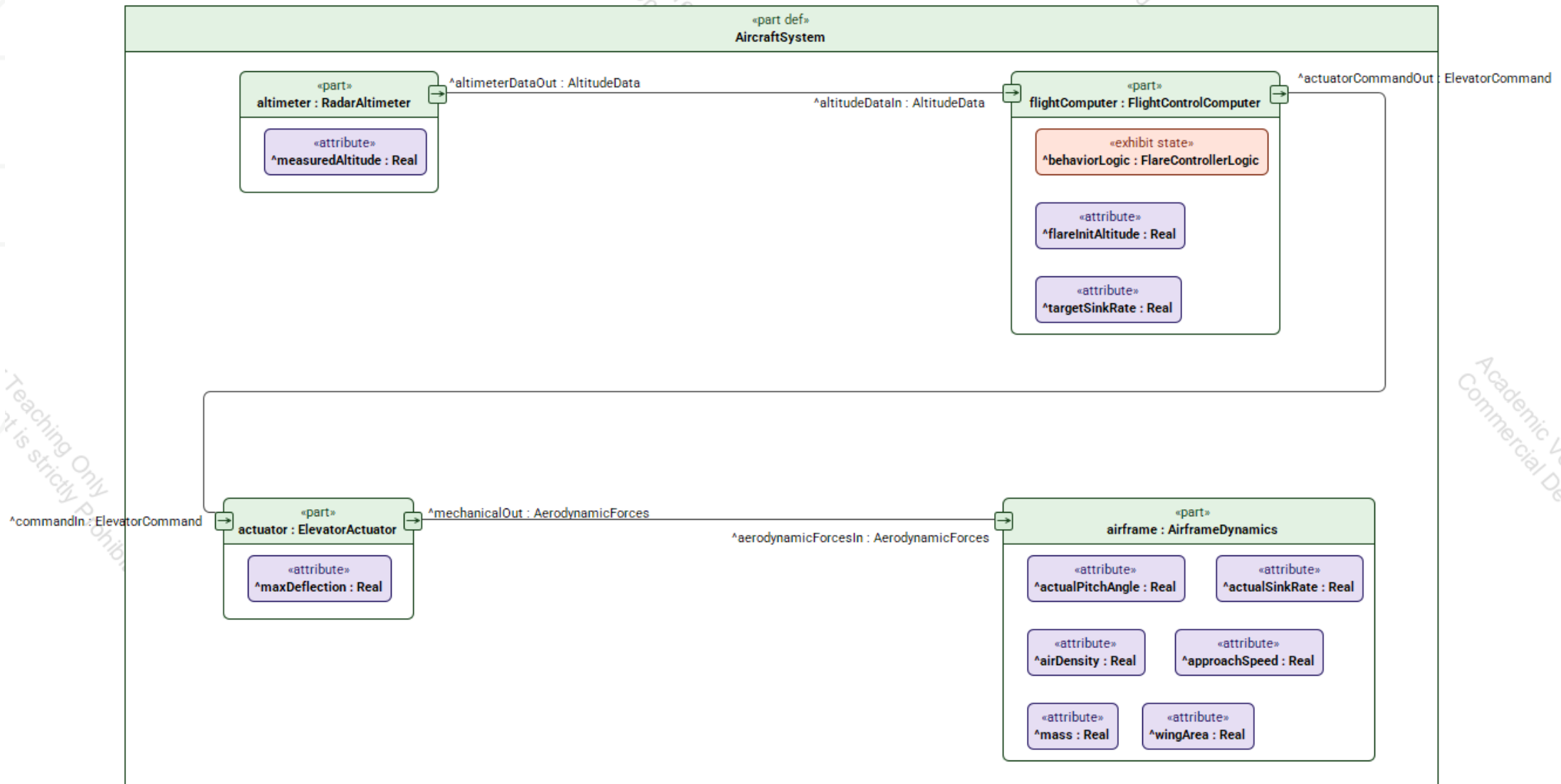


System Behavior

- Initial descent → Landing approach → Flare mode → Touchdown



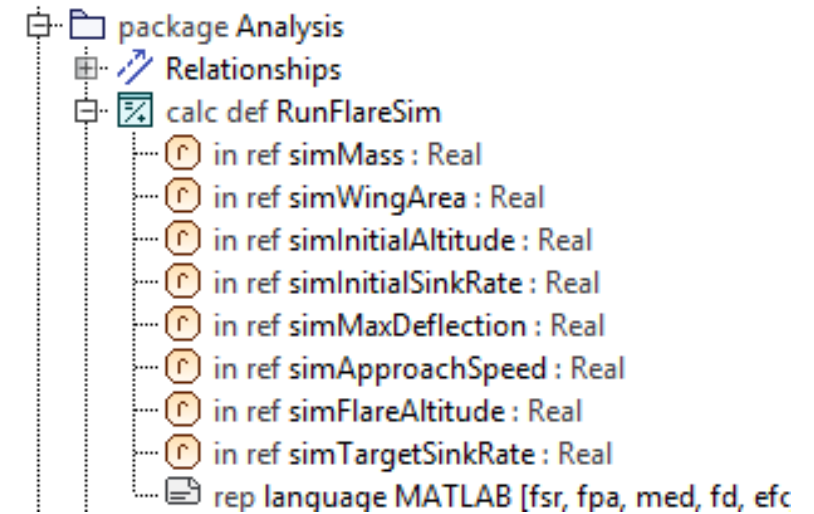
System Structure



Analysis: MATLAB

```
calc def [$144] RunFlareSim {  
  in [$145] simMass : Real;  
  in [$146] simWingArea : Real;  
  in [$147] simInitialAltitude : Real;  
  in [$148] simInitialSinkRate : Real;  
  in [$149] simMaxDeflection : Real;  
  in [$150] simApproachSpeed : Real;  
  in [$151] simFlareAltitude : Real;  
  in [$152] simTargetSinkRate : Real;  
  return [$153] sinkRate : Real;  
  rep [$154] language "MATLAB"  
  /* [fsr, fpa, med, fd, efc, mpr] = runFlareSimulation(simMass,  
  * assignin('base', 'lastFlareResult', struct('fsr',fsr,'fpa',f  
  * sinkRate = fsr;  
  */  
}
```

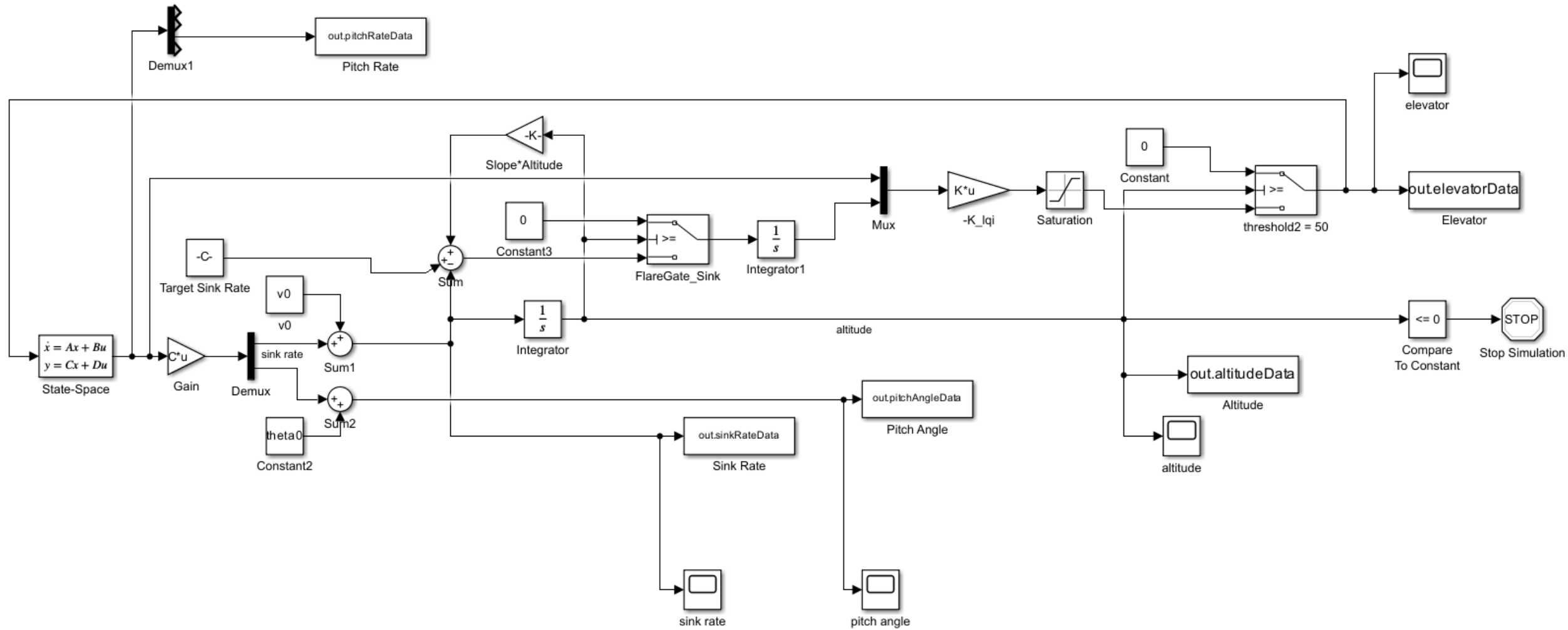
SysML V2 calc definition



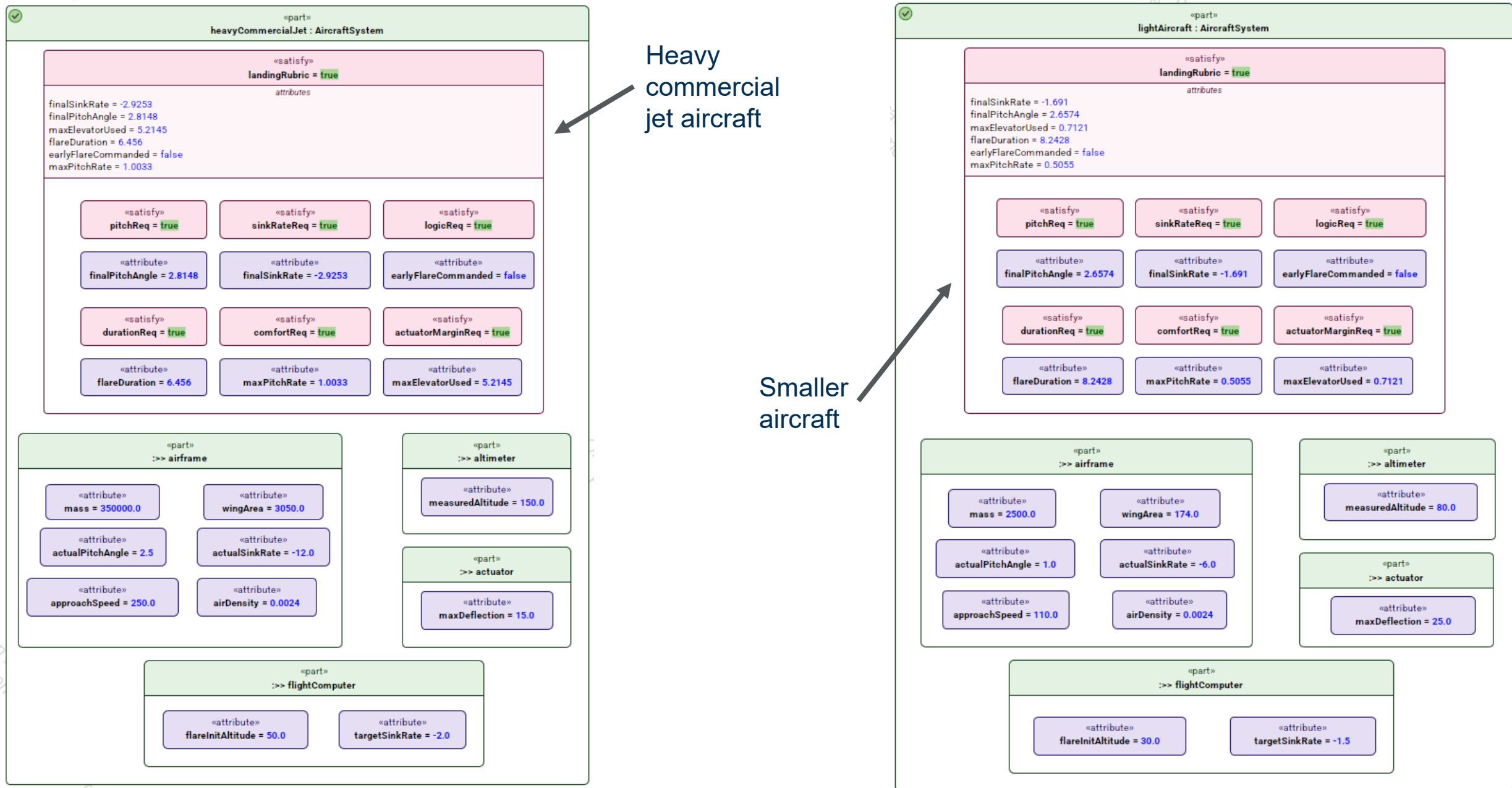
MATLAB Function:

```
function [finalSinkRate, finalPitchAngle, maxElevatorUsed, flareDuration, earlyFlareCommanded, maxPitchRate] = runFlareSimulation(simMass, simWingArea, simI
```

Analysis: Simulink



Instance Evaluation and Verification



Instance Verification Cont.

- Test case fails when the aircraft is made too heavy. Aircraft dynamics can not meet the system requirements.

